

**The Role of Regulators in Mediating Lender Credit Risk, Amidst Accelerating Adoption &
Transaction Volume of Buy Now, Pay Later (BNPL) Applications**

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1. Introduction

The need for consumer credit is inexplicably imminent, given upward trends in GDP growth. In the US, 2025 GDP per capita amounted to \$85,800 (World Economics, 2024) and has been growing by 2.20% over the previous year (World Economics, 2024), with GNI per capita only growing by about 1.6% over the previous year (YCharts, 2024), evidencing a disproportionate increase in the cost of a basket of goods and services per individual. Indeed, White (2025, November 6), a reporter for CNBC, argues that credit card debt has hit an “all-time high of \$930bn” according to data released on February 11 from the Federal Reserve Bank of New York – only 6.4% of such credit card debt representing a fraction of total US debt.

With such large demand for credit, Buy Now, Pay Later (BNPL) applications have garnered significant traction. BNPL applications are defined as alternative payment methods allowing payers to split their payments among several equal parts – most commonly in 4 – over the course of 1-2 weeks. A report from the Consumer Financial Protection Bureau reporting an increase in BNPL loans from \$16.8m in 2019 to \$180m in 2021 (Wang, 2025 January), and “by 2021, around 16% of all users had used BNPL at least once and 30% of these users were persistent users” (Di Maggio et al., 2022, p.2).

2. BNPL: Core Components, Comparison with Credit Cards

Three players of interest form the BNPL ecosystem – first is the customer, second the merchant, and third the BNPL service provider (Akyurek, 2024 July 13). Customers initiate BNPL transactions through purchase of goods and services – often through signing in/up for the BNPL service. The BNPL service conducts an electronic Know Your Customer (eKYC) verification to assess the creditworthiness of the user. Approval of the verification allows the customer to make seamless purchases to merchants without paying the full fee upfront (Akyurek, 2024 July 13) – deferring costs to incur benefits today. BNPL service providers receive their revenue by charging fees to merchants who accept BNPL, and charging “late fees” or “penalties” to customers who fail to pay in a timely manner (Wang, 2025 January).

The key difference between BNPL and credit cards lies in their credit terms. Individuals using credit cards finance their transactions with debt, whilst the credit lender invests their debt –

typically at the risk-free rate (CORRA, SOFR) or a stated Annual Periodic Rate (APR). BNPL, on the other hand, is structured to incur fixed installment loans “with a down payment due at sale,” (Wang, 2025 January). Indeed, BNPL loans do not possess any additional fees and interest, limited credit checks or credit reporting, and are offered by retailers – not the service provider itself.

The looser lending terms offered by BNPL attract highly price-sensitive consumers. These consumers are often found in highly price-elastic markets, where such pricing models like surge pricing, penetration pricing via steep discounts, and price skimming have great effects on the customer's willingness to spend and their expected utility from the received good/service. In addition, BNPL providers often charge merchants higher fees than credit card companies (Wang, 2025 January) – thusforth incentivizing merchants with BNPL services to hike up prices to reach market equilibrium. Such an action is done with the motive to rationally maximize efficiency, but behavioral economic theory – commonly the work of Kahneman & Tversky, shows that customers make irrational decisions in the pursuit of profit, often through behavioral biases such as the availability effect, or present bias. Because customers are irrational, they may tend to overvalue the discount provided by BNPL and increase their debt exposure – which, if they later find difficulty servicing – can heighten risk and costs to lenders who seek stable cash flows.

3. How Lenders Assess Credit Risk: BNPL vs. Traditional Credit

For auto, mortgage, and credit card loans, lenders commonly use a five-fold framework referred to as the “5 Cs of Credit” to determine the creditworthiness of an individual. The first, **Character**, details the individual’s credit history, which is often formally reported in credit reports. **Capacity** refers to the overall ability to pay loans – recorded through debt-to-income (DTI) ratios, with lower DTIs preferred. **Capital** refers to assets/investments, such as down payments, willing to be put up at the date of settlement for servicing the loan. **Collateral** are secured liens (ownership by lenders of particular assets/investments) that a lender may seize in the event of default – for which a borrower fails to meet his payment obligations. Finally, **Conditions** – such as the specific purpose of the loan, as well as external macroeconomic factors such as monetary policy & industry trends influence lending terms (Capital One, 2023 July 27).

BNPL introduces several caveats to this five-fold framework. As “lenders generally do not furnish loan performance information to the three Nationwide Consumer Reporting Companies (NCRC), it has not been possible to use administrative records to estimate BNPL prevalence, persistence in use, and loan stacking across firms” (CFBP, 2025 January). This manifests into what is commonly known as “phantom debt” – debt concealed by individuals from lenders, to make them appear more creditworthy than they really are. Krause (2025) details the action of “loan stacking” by borrowers – for which multiple BNPL loans could be carried simultaneously across different providers – resulting in negligible efforts by lenders to assess a user’s debt capacity. Indeed, both price sensitivity and younger demographics, specifically between 14-29 years old, are instrumental in understanding this issue.

4. Market Demographics: BNPL

Younger demographics, characterized by their higher willingness to take risk, but low ability to follow through and take risks, possessed a “76% higher delinquency rate” compared to other demographics. This trend has only grown, with Van Der Klaauw of the New York Fed reporting that “delinquency among credit card borrowers have steadily risen since 2016, notably among younger borrowers” (White, 2025 November 6). With respect to the Diffusion of Innovation curve - which details the order by which individuals of all demographics adopt a new innovative technology or product, a paper done by Dedehayir et al. (2017) finds that while early adopters and age are not directly correlated, “early adopters are deemed to be opinion leaders” (p.3) - whom typically disseminate new innovations to younger demographics much faster than any other demographic via social media channels – as well as spark FOMO and herding bias to assist conversion. These individuals often adopt innovations first to exploit their relative advantages and substitutes without considering risk factors. The exploitation of BNPL’s advantages through “loan stacking”, which are not adequately risk-managed compared to substitutes, facilitates the accumulation of phantom debt – generating significant concerns among lenders and justifying higher risk premiums to compensate for opaque risk.

A paper by Brewer & Arber (2025, February 25) additionally suggests that BNPL providers conduct external credit risk mechanisms that facilitate their underwriting process. “BNPL companies will sometimes assess ‘more than one billion data points’ to determine how likely a

person is to pay back their loan – including, but not limited to, the rate of payback on the specific item they are purchasing, a LinkedIn profile indicating the kinds of jobs the consumer has worked, average incomes for the zip code that the item is being shipped to, and other vague indicators. However, the ambitious concept of analyzing billions of data points to assess credit risk introduces a variety of risks. Large collections of personal, public data leads to inaccurate credit risk scoring relative to the fundamentals used by credit card lenders. Any efforts to collect more personal data may violate privacy law – for example, the California Privacy Rights Act, for which “consumers have the right to request that businesses delete any personal information collected about them” (Consumer Privacy Act, n.d.) if the consumer may be unwilling for third parties to collect data from them, even if that information is instrumental by BNPL providers for stable user cash flows. Finally, data bias unfairly charges particular groups of individuals higher rates, however the lack of transparency behind these algorithms make it difficult to pinpoint sources of bias, or scrutinize BNPL providers’ proliferation of rate discrimination (Brewer & Arber, 2025 February 25).

5. Regulators Response

Major regulatory bodies have stepped forward to voice concerns, with actions from the Consumer Protection Bureau to clarify that “Buy Now, Pay Later lenders are ‘credit cards’ under Regulation Z” (Consumer Financial Protection Bureau, 2025, p.6), and New York enacting “legislation that will require providers of BNPL loans to be licensed by the state and adhere to various disclosure and practice requirements” (Pearson & Mitzenmacher, 2025 June 6). Still, appropriate safeguards may not yet be in place to truly mitigate credit risks from mass BNPL adoption. The interpretive rule received pushback from industry stakeholders, who argued, “the structural and economic features of BNPL products were distinct from credit cards and ill-suited to existing credit card regulation” (Krause, 2025, p.3), and therefore, resulted in a revocation of the rule in March 2025. Indeed, because the rule has been made interpretive, it is in no position to receive considerable attention from federal legislators. Furthermore, BNPL lenders do not report or have the same capabilities as accredited financial institutions in reporting loan performance information to credit reporting facilities. Thereby, weakening estimation of credit risks emerge, and information asymmetry between borrowers and lenders facilitate overleveraging (taking on too much credit) and the risk of borrower fraud. BNPL regulation remains flawed, but sought after - with regulators

motivated to maintain the integrity of the financial system, and prevent recurrence of a systemic collapse akin to the Global Financial Crisis (GFC) in 2008.

6. Losses to Lenders

Unregulated credit risks from BNPL consumer lending pose major risks to lenders if borrowing is mismanaged. For instance, excessive lending to subprime borrowers - consisting of younger, or lower-income individuals who take on excessive risk or exploit credit loopholes - raises default rates among investors to compensate for increased risk. Indeed, the Consumer Financial Protection Bureau (2025) finds that “default rates are highest amongst the youngest borrowers, those between 18 and 24” (p.20); however, these individuals also comprise the largest share of BNPL borrowers at 37.1% (p.25). The additional lack of credit scoring features contributes to higher rates, as BNPL lenders suffer from information asymmetry and cannot price risk to borrower creditworthiness. From the investor’s perspective, higher default risks lower risk-adjusted returns, as lack of regulation leads to less transparency, greater uncertainty regarding defaults, and volatile cash flows. As BNPL regulations continue to evolve, emphasis on promoting transparency, maintaining stable cash flows, and ensuring adequate risk diversification helps protect the stability of lenders’ returns while supporting the long-term, sustainable growth of BNPL platforms.

7. Methodology

For our methodology of regulatory intervention, we will provide data for two major hypotheses: first, we will test whether an increase in BNPL network effects increases accumulated phantom debt. Second, we will test whether an increase in accumulated phantom debt resulted in increased expected lender losses, with the key variables being Probability of Default (POD), Loss Given Default (LGD), and Exposure at Default (EAD). Lenders in this case refer to the BNPL counterparty - typically retailers, but also includes BNPL service providers – each party differing from their EAD. Third, we will look at key variables that produce a countereffect to increases in accumulated phantom debt and expected lender losses, which serve as robust regulatory intervention opportunities. We will finally model this outcome at scale, looking primarily at the

2008 Global Financial Crisis, where similar phantom debt accumulated through the issuance of high-risk subprime mortgage loans to individuals with poor creditworthiness, thus leading to systemic credit losses.

8. Effects of BNPL on Phantom Debt

According to Cornelli et al. (2023, December 4), “BNPL users in the United States are more highly indebted and have lower credit scores than non-users” (p.68), indicating that BNPL demand is concentrated among borrowers already facing constrained liquidity. This issue was exacerbated following the halt of government stimulus packages at the end of the COVID-19 pandemic, where “failed BNPL loan payments spiked significantly, whereas credit card loan failures did not,” resulting in a 4 times higher than the latter delinquency rate.

Indeed, Cornelli et al. (2023, December 4) conducted a regression analysis, involving the use of key macroeconomic variables to examine expanding BNPL adoption’s influence on accumulated phantom debt. Inflation has a strong positive elasticity with BNPL use ($\beta \approx 0.40\text{--}0.69$) (p.70-71), showing that decreases in real purchasing power results in increased deferral of payments and fragmented debt. E-commerce development was found to be the strongest predictor of BNPL uptake ($\beta \approx 1.23\text{--}1.84$) (p.70-71). Moreover, countries with loose financial regulation and weaker consumer protection – proxied by lower regulatory quality scores and lower banking-stringency indices – show significantly higher BNPL usage ($\gamma \approx -3.24$ to -3.36). Banking-sector inefficiency also increases BNPL activity, with higher bank cost-to-income ratios ($\delta \approx 4.13\text{--}4.23$), alongside greater household indebtedness ($\theta \approx 0.38$). These variables together conclude that expanding economic activity with insufficient financial regulation facilitates the accumulation of phantom debt across BNPL contracts & providers. Demyanyk and Van Hermert (2008) detail that “lending booms” are often associated with a “banking crisis,” with the major stimulus being “changes in information asymmetry across banks”, and “low screening activity” leading to “intense price competition, lower standards, and lower profits” (p.3).

9. Phantom Debt effect on Lender Losses

Expected loss to credit investors is commonly modelled in the form of Expected Loss (EL) = (PD) * (LGD) * (EAD). Under the Basel internal-ratings-based framework, PD is defined as the one-

year probability by which credit investors may not receive their payment (BIS, 2023 January 1), LGD is the share of exposure that is not recovered in a credit loss, and EAD is the outstanding exposure at the time of default (BIS, 2023 January 1).

The accumulated phantom debt on BNPL loans affects lender losses primarily through raising the true PD relative to what is implied by observable bureau scores and income metrics. BNPL credit is extended on the basis of “soft” checks – credit checks not done with sufficient due diligence, is often not reported to credit bureaus, and late payments are only partially visible, so a material portion of a borrower’s BNPL debt remains outside traditional credit data (Cornelli et al., 2023, p.63). A report by the CFPB (2025) shows that of total borrowers, significant portions of BNPL loans go to subprime (17.6%) or deep-subprime (46.8%) borrowers (p.30), that lenders approve roughly 78% of applications from such borrowers (p.13), Phantom debt is this stack of lightly underwritten, weakly reported BNPL obligations: leaving lenders unaware of the borrower’s full leverage, causing realized PDs on BNPL portfolios – and even on other unsecured credit held by the same borrowers – to be higher than estimates based only on bureau data.

The impact on EAD differs between merchants and BNPL service providers. In the standard merchant-partner model, once a BNPL transaction is approved, the platform pays the merchant the full purchase price upfront and assumes the customer’s credit risk; merchants mainly face higher acceptance fees but shift credit and fraud risk to the platform. Their effective EAD to consumer default is therefore close to zero, aside from operational disputes or chargebacks. By contrast, the BNPL lender books an exposure equal to the financed share of the purchase price (typically 75% of the ticket value after a 25% down payment) plus any capitalized fees.

Furthermore, LGD in this setting is largely a function of how much funds – typically equity and secured assets (collateral) a borrower has posted through down payments, and how recoverable the financed asset is. BNPL loans are unsecured; the primary loss-mitigating features are the initial payment at checkout, subsequent automatic debits, and late-fee/collection mechanisms. In the common “Pay In 4” model, a 25% down payment is due at purchase, with three further equal payments over 6 weeks, so the lender’s EAD at inception is 75% of the purchase price. Longer-tenor BNPL or “pay monthly” products accept smaller or even zero down payments, effectively

increasing the loan-to-value ratio of each transaction and raising LGD. Phantom debt continues to accumulate via repeated small down payments and growing stacks of payment obligations – which reduces present borrower funds, while lenders’ outstanding financed portions expand.

10. Regulatory Intervention Opportunities: Effects with Negative Correlations on Phantom Debt & Lender Losses

BNPL mostly proliferates, and its frequency of abuse is much heavier in jurisdictions with looser financial and consumer-protection regulation; higher values of banking-stringency and regulatory-quality indices. These variables exist to increase the frequency of phantom debt in the credit markets – however, the opposite effect is true when these variables are subsequently decreased. The decrease in these variables presents an opportunity cost, which forgoes the growth prospects of BNPL for decreasing accumulated phantom credit. Regulators must intervene where the costs of one opportunity (growth prospects of BNPL) outweigh its benefits. Several solutions can be addressed given the variables above.

The core solutions that drive the mitigation of accumulating phantom debt is increasing the comprehensiveness of credit reporting, minimum affordability tests, limits on concurrent BNPL contracts, requirement of a larger down payment upfront for payment charges, and adequate guidance to banks on assisting in BNPL debt exposures. Comprehensive credit reporting reduces the information asymmetry between lenders and borrowers, which allows lenders to obtain a better understanding of a borrower’s creditworthiness. A greater understanding of a borrower’s creditworthiness reduces their PD – reducing the expected loss lenders may incur, and providing optionality to diversify higher-risk borrowers using credit default swaps & credit limits. Additionally, credit limits, such as minimum affordability tests, limits on amounts of loans, and greater down payments upfront allow BNPL providers to adequately discipline delinquent users to manage their loans with care in their ability to timely service them. Credit limits vary by reducing the LGD or EAD – for minimum affordability tests & loan limits, BNPL providers effectively limit their default exposure (EAD) to phantom debt & benefit their shareholders by maintaining stable, fairly valued cash flows. Furthermore, given that LGD is the inverse of the Recovery Rate (RR), greater upfront down payments increase the present value of the transaction and effectively the amount recovered, resulting in a lower sensitivity of rate changes to default risk. Finally, adequate

guidance to banks, through considering the usage of alternative financing platforms in lending debt, or handling bankruptcy proceedings, improves the preparedness of banks to provide actuarial services to BNPL providers in the case of mass defaults. This allows BNPL providers to grow their business model, innovate, and continue to provide alternative credit financing for less creditworthy individuals, such as younger demographics and those avoiding revolving credit. In summary, our findings suggest that jurisdictions and periods that have implemented these solutions are expected to show lower BNPL usage, lower delinquency rates, and smaller realized LGDs due to earlier intervention and better loss-mitigation options.

11. Applied at Scale: 2008 Global Financial Crisis & Subsequent Regulatory Absolvment

To apply this framework at scale, we treat the 2008 Global Financial Crisis as a historical analogue of phantom-debt buildup in a different asset class: subprime and non-traditional mortgages securitized into mortgage-backed securities (MBS) and collateralized debt obligations (CDO). Demyanyk & Van Hemert (2008) detailed “progressively weakening underwriting standards, rising loan-to-value ratios, and the commonness of complex features such as teaser rates and negative amortization”, all of which amplified default risk but were not fully reflected in risk assessments of structured products (p.3). Regulators later characterized this as a breakdown in both market discipline and oversight of unregulated or lightly regulated credit markets, with Felton & Reinhart (2008), along with Demyanyk & Van Hemert (2008), purporting that weak underwriting, opaque securitization structures, and mispriced risk lead to large, correlated losses following the collapse of the mortgage markets.

Actions in response to regulating the 2008 Global Financial Crisis serves direct parallels to the solutions for regulating BNPL applications. In retail mortgage markets, Hizmo & Sherlund (2018, November 16) detail the Dodd-Frank Act’s Ability-to-Repay and Qualified Mortgage rules requiring lenders to verify income, debt obligations and repayment capacity, and deem QM loans to be lower-risk products that avoid many of the pre-crisis high-risk features. At the system level, the Basel III framework raised the quantity and quality of bank capital, introduced leverage ratios (ratios detailing the acceptable level of debt) and capital buffers (required reserves held by financial institutions), and added liquidity standards such as the Liquidity Coverage Ratio to limit

excessive changes in BNPL payment maturity (Bank for International Settlements, 2017, p.1). IOSCO and other standard setters have also cited “providing regulatory support for improvements in disclosure by issuers to investors,” “review investor suitability requirements”, and “encourage the development of tools by investors to assist in understanding complex financial products,” (International Organizations of Securities Commissions, 2009, p.26). This mirrors New York’s BNPL licensing law discussed above, where BNPL lenders must meet consumer-protection, reporting, and supervisory requirements similar to those applied by other credit providers (Pearson & Mitzenmacher, 2025).

Thusfar, these reforms, initiated as a response to the 2008 GFC, serve as compelling evidence targeting regulation, improved information, and higher loss-absorbing capacity that materially reduced the system-wide consequences of phantom-debt build-ups. Credit-bureau integration, affordability rules, capital guidance and consumer-protection standards—dampen the BNPL cycle by lowering both the stock of hidden obligations and the EL borne by lenders and investors.

12. Conclusion: Results & Discussion

To conclude, the increased adoption of BNPL accelerates the transmission of phantom debt through the credit markets. Cornelli et al.’s regression model on phantom-debt proxies showed positive and economically meaningful coefficients. The effect was strongest in environments with high inflation and deep e-commerce penetration, for which BNPL lenders obtain the most revenue solely from transaction volume. However, transaction volume does not imply cash flow stability, and because BNPL users are more indebted, have lower credit scores, and experience delinquency spikes when financial aid like COVID-19 government stimulus packages are withdrawn, market overvaluation will be met with a significantly large loss. It’s worth noting that market beta – the total undiversifiable & unavoidable risk of the market – is higher when the correlation between the market is high. Indeed, increased volatility, denoted as the standard deviation of the market and the security (in this case, the BNPL counterparty, the borrowers for which cash flows are expected) also increases beta, and larger BNPL transaction volume results in greater uncertainty in lender’s receivable cash flows.

Linking phantom debt to lender outcomes through the $EL = PD \times LGD \times EAD$ formula, we also find that phantom-debt indicators are significant predictors of BNPL delinquency rates. This supports the view that soft checks and incomplete credit reporting result in an underestimation of PD. Portfolios with longer maturities or lower down-payment structures exhibit larger LGDs, while borrowers with multiple overlapping plans drive higher EAD on BNPL balance sheets relative to merchants.

Regulatory and institutional actions exhibit considerable success in mitigating these effects. Higher banking-stringency and regulatory-quality scores, BNPL-specific conduct rules, and mandatory bureau reporting are associated with lower BNPL usage among already highly indebted households, and with smaller marginal increases in PD and EL for a given BNPL intensity. Viewed alongside the 2008 Global Financial Crisis—where weak underwriting and opacity in subprime securitization produced large, correlated losses—our findings suggest that early application of analogous tools (affordability tests, capital guidance, comprehensive reporting, and robust consumer protection) can materially limit phantom-debt accumulation and its transmission into lender losses.

13. Future Opportunities:

As innovation never ceases, our paper hopes to discuss numerous evolving trends in the BNPL space that regulators must consider. Web 3.0 is considerably rising in popularity, accompanying a cultural shift to decentralized ecosystems that center on user autonomy and self-sovereignty, with cryptocurrencies like stablecoins & altcoins being frequently used in virtual worlds & marketplaces to facilitate transactions. BNPL is expected to play an instrumental role, potentially in improving liquidity or hedging volatile price movements for digital securities, through derivative swaps. Additionally, similar to the Global Financial Crisis, we'd like to explore the possibility of BNPL asset-backed securities, and key regulatory actions from bodies like the SEC (Securities & Exchange Commission) & FINRA (Financial Industry Regulatory Authority) in legitimizing the securitization of such investments. Finally, we'd like to explore key trends in Artificial Intelligence (AI) in powering quantitative credit scoring models, allowing BNPL providers more efficient means of assessing user credit, albeit with prudence regarding privacy & security laws & best practices. Further research will explore such credit innovations in greater

detail, it's risks, and proactive regulatory responses from precedent events such as the Global Financial Crisis, and cryptocurrency market rugpulls & crashes to ensure the economic stability of the fintech system.

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Appendix:

$$\ln(BNPL_{it}) = \alpha \ln(Inflation_{it}) + \beta \ln(Ecommerce_{it}) + \gamma Regulation_{it} \\ + \delta \ln(Bank\ cost\ to\ income_{it}) \\ + \theta Household\ debt_{it} + Other\ controls_{it} + FE + \varepsilon_{it}$$

Drivers of BNPL usage¹

Table 2

	Ln(BNPL users per million people)			
	(I)	(II)	(III)	(IV)
Ln(Inflation)	0.398** (0.165)	0.557** (0.205)	0.688*** (0.228)	0.685*** (0.226)
Ln(E-commerce, % of total sales)	1.225*** (0.187)	1.810*** (0.262)	1.728*** (0.313)	1.840*** (0.334)
Index of banking stringency ²		−34.769*** (3.300)	−37.580*** (2.972)	−37.398*** (3.470)
Regulatory quality index ³		−3.242*** (0.329)	−3.355*** (0.358)	−3.363*** (0.410)
Ln(Bank cost-to-income)			4.225*** (0.918)	4.132*** (0.906)
Household debt to GDP			0.381* (0.190)	0.377** (0.183)
Other controls ⁴				✓
Number of observations	378	373	361	359
Adjusted R-squared	0.895	0.909	0.909	0.910

¹ The sample covers 25 countries for the period Q2 2016–Q4 2022. All the specifications include country and time fixed effects. Standard errors clustered by time in brackets; ***/**/* indicates statistical significance at the 1/5/10% level. See technical annex for details.

Sources: Cornelli et al (2023); BIS credit statistics; IMF, *World Economic Outlook*; UNU-WIDER, *World Income Inequality Database (WIID) Companion dataset*; eMarketer Inc; Sensor Tower; World Bank; authors' calculations.

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